A Timeline of Artificial Intelligence

From Turing's question to generative AI

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THE BIG PICTURE

Six eras of AI, with booms and winters

1943-1973 Birth of AI	1974-1980 Winter 1	1980-1986 Expert systems	1987-1993 Winter 2	1993-2011 Data and chess	2012-now Deep learning
Turing Test, Dartmouth, perceptron, ELIZA	Funding cuts after unmet promises	Rule-based AI sells to business	Lisp market collapses, projects fall short	Deep Blue, cloud, GPUs, ImageNet	AlexNet, GANs, transformers, ChatGPT

Pattern: hype, funding boom, unmet promises, winter. Then hardware and data caught up with the math.

Foundations: the idea of thinking machines

1950

Turing asks if machines think

Alan Turing proposes the imitation game, later called the Turing Test, to judge machine intelligence.

1956

Dartmouth names the field

John McCarthy, Marvin Minsky, Claude Shannon and others meet. The term artificial intelligence is coined.

1958

The perceptron

Frank Rosenblatt builds a machine that learns by adjusting weights, an ancestor of today's neural networks.

1969

A book exposes limits

Minsky and Papert show single-layer perceptrons cannot solve some simple problems. Neural net interest cools.

AI WINTER

First AI winter, 1974 to 1980

1974-1980

Trigger: the 1973 Lighthill Report in the UK and funding cuts by DARPA in the US.

Causes

- Big promises, such as machine translation, went unmet
- Computers had too little speed and memory
- Government sponsors lost patience with open-ended research

Impacts

- Research money dried up in the UK and US
- Work shifted to narrow, practical problems
- Neural network research slowed after the Perceptrons book

The expert systems boom

1980

XCON configures computers

Digital Equipment Corporation uses a rule-based system to fill orders, an early sign AI could pay for itself.

1982

Japan's Fifth Generation

Japan funds intelligent computers. The US and UK respond by restarting their own AI funding.

1986

Backpropagation spreads

Rumelhart, Hinton and Williams show how multi-layer networks learn, reviving neural networks.

1989

Neural nets read zip codes

Yann LeCun's team trains a convolutional network to read handwritten digits on mail.

AI WINTER

Second AI winter, 1987 to 1993

1987-1993

Trigger: the collapse of the Lisp machine market in 1987.

Causes

- Cheap desktop computers beat costly specialist AI hardware
- Expert systems were hard to update and broke outside their rules
- Big projects, like Fifth Generation, fell short of goals

Impacts

- Many AI companies failed and investors pulled back
- Some researchers avoided the AI label
- Steady work on statistics and neural nets set up the revival

Chess, cloud, GPUs and big data

1997

Deep Blue beats Kasparov

IBM's chess computer defeats the world champion by searching moves fast, not by learning.

2006

Cloud computing arrives

Amazon Web Services launches EC2, so anyone can rent computing power by the hour.

2007

GPUs open up to AI

NVIDIA releases CUDA, letting graphics chips run the parallel math neural networks need.

2009

ImageNet supplies data

Fei-Fei Li's team releases millions of labeled images, the large dataset deep learning needed.

2012-2022 | DEEP LEARNING TO GENERATIVE AI

From AlexNet to ChatGPT

2012

AlexNet wins ImageNet

Krizhevsky, Sutskever and Hinton train a deep network on GPUs and sharply cut image errors.

2016

AlphaGo beats Lee Sedol

Google DeepMind's AlphaGo wins 4 games to 1 in Go, long seen as too complex for computers.

2017

Transformers arrive

Google researchers publish Attention Is All You Need, the design behind modern language models.

2022

ChatGPT goes public

Built on OpenAI's GPT models, it arrives with DALL-E 2, Imagen and PaLM, reaching millions.

Computing power exploded

AlexNet (2012) used 470 petaFLOP. PaLM (2022) used about 2.5 billion, roughly 5 million times more.

Leading companies and models

OpenAI: ChatGPT, GPT models, DALL-E

Google DeepMind: Transformer, AlphaGo, Gemini

Anthropic: Claude models

Meta: Llama open-weight models

NVIDIA: GPUs and CUDA for training

Cloud: AWS, Azure, Google Cloud rent compute

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AI disclosure: Claude was used to help gather and organize milestones. I reviewed and edited the document.